
Working Memory in the Chain of Thought Is Not Captured by J-Space

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Abstract

1 A common reading of J-space, the concept workspace a fitted Jacobian lens reads
2 from the residual stream, is that reasoning state is carried internally and the chain
3 of thought externalizes state the workspace can no longer hold. Behavioral mea-
4 surements from 1.5B to 671B parameters show no sign of such relief. Instructed to
5 answer chained arithmetic without writing, models fail beyond one to two depen-
6 dent steps. In free generation, models write 0.93 to 1.00 of intermediate values
7 from one-step problems on, so writing has no onset threshold. No intervention
8 moves state across the boundary: damaging the residual stream does not increase
9 writing, capping writing does not move computation inward, filler tokens in place
10 of real writing do not restore accuracy, and ablating the top active J-space concepts
11 changes nothing against a size-matched random control at any dose up to full pro-
12 jection removal. We then locate the memory that does carry the state, with causal
13 interventions on six unmodified models from four families (4B to 14B). The hid-
14 den state under a written number holds the value later steps use. Overwriting it
15 with the state a different number would have produced, text unchanged, redirects
16 the final answer to that never-written number in 0.70 to 1.00 of runs; size-matched
17 random noise redirects at most 0.01 on the four instruction-tuned models. Later
18 steps retrieve the value through attention to its position, the retrieval completing
19 by layer 23 of 36 on Qwen3-4B. Two written positions hold values independently,
20 and planted pairs compose into a sum appearing nowhere in the input (0.88 and
21 0.90). Identical text can therefore accompany different stored values, and a moni-
22 tor that reads only the text cannot tell them apart.

23 1 Introduction

24 The chain of thought is the primary working memory of serial reasoning in language models: denied
25 writing, models fail within one to two dependent steps, and no intervention we test shifts state
26 between the internal workspace and the written trace. The behavioral claims are established at 1.5B
27 to 671B parameters; the mechanism, with causal interventions on six unmodified models from four
28 families at 4B to 14B, on two task families with exact ground truth for every intermediate value.

29 The capacity half addresses the overflow picture of externalization: reasoning state is carried inter-
30 nally, and writing takes over when the workspace can no longer hold it. This is a natural reading of
31 workspace accounts of reasoning, including J-space, the concept workspace a fitted Jacobian lens
32 reads from the residual stream [Gurnee et al., 2026]. A workspace that writing relieves should show
33 up in three places; it shows up in none.

34 1. **No fallback** (behavioral, 1.5B to 671B). Instructed to answer chained arithmetic with no
35 writing, models fail beyond one to two dependent steps (Section 3).

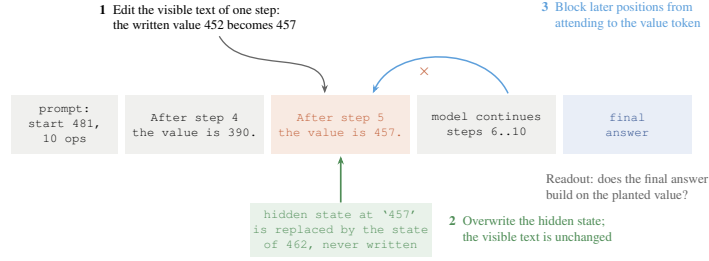


Figure 1: The edit-and-continue protocol. One written step of a correct worked solution is altered and the model continues from it; the readout is whether the final answer builds on the altered value. The interventions: (1) edit the visible text of the step; (2) overwrite the hidden state at the value’s token, leaving the text unchanged; (3) block attention from later positions to the value’s token.

- 36 2. **No threshold** (behavioral, 1.5B to 671B). Models write nearly all ground-truth intermedi-
37 ates from one-step problems onward (Section 3).
- 38 3. **No reallocation** (behavioral and ablation, 4B to 7B). Damaging the residual stream does
39 not increase writing, capping writing does not move computation inward, filler tokens do
40 not substitute for real writing, and the J-space ablation changes nothing at any dose up to
41 full projection removal (Section 3).
- 42 4. **Storage** (causal, 4B to 14B). The hidden state under a written number carries the value the
43 model later uses: planting a never-written value there, text unchanged, redirects the answer
44 on all six models (Section 4).
- 45 5. **Retrieval** (causal, 4B to 14B). Later steps read the value through attention to the written
46 number, in the middle third of the network’s layers (Section 5).
- 47 6. **Composition and trust** (causal and behavioral). Two positions are independently address-
48 able, and planted values compose into their sum. Reliance on the trace tracks whether the
49 model can regenerate the state, a candidate explanation for conflicting reports on how trace
50 reliance scales (Sections 4 and 6).

51 Prior work anticipated the memory role of written tokens theoretically [Merrill and Sabharwal,
52 2024], behaviorally [Lanchantin et al., 2023], and causally on a fine-tuned model [Shih et al., 2026]
53 and on multiplication and dynamic-programming tasks [Zhu et al., 2025], without testing whether
54 an internal alternative exists, how the value is retrieved, or whether stores compose.

55 2 Experimental setup

56 2.1 Tasks and models

57 Both task families give exact ground truth for every intermediate value, so the downstream effect
58 of any edit is computable. Arithmetic chains start from a three-digit number and apply d signed
59 two-digit additions and subtractions. Narrative state-tracking problems describe a character whose
60 count of an item changes through varied events and surface forms. In both, the model writes one
61 line per step giving the running value. A third task, box tracking, updates the contents of several
62 containers in parallel; it appears only as the internally held comparison task for the lesion test in
63 Section 3. White-box experiments use Qwen3-4B, Qwen2.5-7B-Instruct, Phi-3-medium, OLMo-2-
64 7B-Instruct, and DeepSeek-R1-Distill-Qwen 7B and 14B; the 1.5B distill appears in the behavioral
65 scale range only. Frontier models (DeepSeek V3.2, DeepSeek-R1-671B, Claude Sonnet 4.5, Llama
66 3.3 70B) are tested behaviorally through assistant prefill.

67 2.2 Edit-and-continue protocol

68 The protocol takes a correct worked solution, changes the value at one step j , truncates immediately
69 after the edited line, and lets the model continue (Figure 1). A no-edit truncation measures the
70 sampling noise floor, 0.00 throughout. Generation uses temperature 0.6 and top- p 0.95 with five

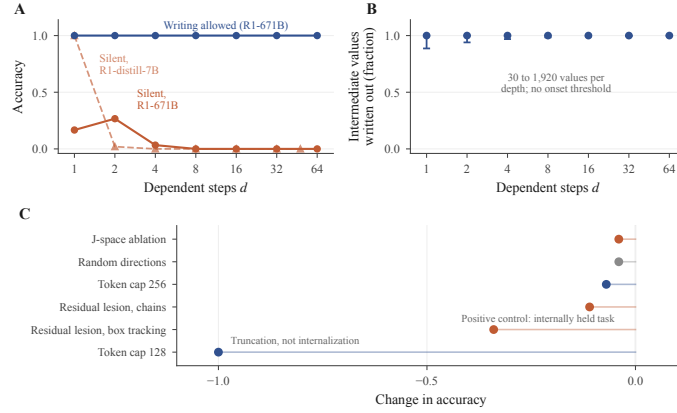


Figure 2: Writing is not triggered by exhausted internal capacity. (A) Instructed to answer without writing, models collapse after one to two dependent steps; with writing allowed, accuracy stays near ceiling ($n=30$ items per depth). (B) The fraction of intermediate values written is 0.93 to 1.00 at every depth (Wilson intervals); no onset threshold appears. (C) No intervention moves state between the workspace and the trace: J-space ablation and random directions leave accuracy unchanged; residual lesions hurt the internally held box-tracking task far more than the written chain task; a token cap below the per-step floor collapses accuracy by truncation.

71 samples per item; unparseable outputs are counted separately. Intervals are 95 percent bootstrap
72 intervals over items. Appendix C summarizes generation and intervention configuration; prompts,
73 model revisions, and per-run data will be released with the code.

74 3 Does writing replace internal working memory?

75 3.1 Performance without written intermediates

76 Instructed to output only the final answer (Figure 2A), with token counts confirming nothing else
77 was generated, models fail beyond one to two dependent steps (Qwen2.5-7B: 1.00 at $d = 1$, 0.04 at
78 $d = 4$; DeepSeek-R1-671B: at most 0.27 at any depth, 0.03 at $d = 4$; DeepSeek V3.2 on modular-
79 arithmetic chains: 0.23 at $d = 1$, at most 0.03 beyond). Prefilling the answer turn with inert dot
80 tokens matched to the twelve-token-per-step budget of real traces [Pfau et al., 2024] does not rescue
81 the collapse: at $d = 4$, 0.13 with filler against 0.07 without on Qwen2.5-7B and 0.07 against 0.00
82 on Qwen3-4B, and 0.00 for both at $d = 8$ ($n = 30$ per cell). Qwen3-4B alone recovers at $d = 2$
83 (0.40 to 0.97), the parallel-width benefit filler tokens are known to provide, which does not extend to
84 the depths where writing is needed. The values are random, so nothing can be recalled; the failures
85 bound performance under this instruction, with its prompting and training-distribution effects, rather
86 than latent internal capacity.

87 3.2 When models write intermediate values

88 Models write 0.93 to 1.00 of ground-truth intermediates in free generation from $d = 1$ through
89 $d = 64$, at every scale tested: no difficulty threshold at which writing begins is observed. This
90 rules out an inference-time switch from internal holding to writing; an always-write policy learned
91 in training could itself descend from capacity limits.

92 3.3 Tests for reallocation between internal and written state

93 A residual-stream lesion at fixed dose lowers accuracy on box tracking, whose state is held inter-
94 nally, by 0.34, and on the written chain task by 0.11; a lesion strong enough to cross the accuracy
95 cliff makes traces disintegrate rather than expand. Hard token budgets compress wording up to 2.5
96 times while keeping 0.98 to 1.00 of values, then truncate below roughly twelve tokens per step. Ab-
97 lating the top 16 active J-space concepts at every position, at the strongest dose preserving ordinary

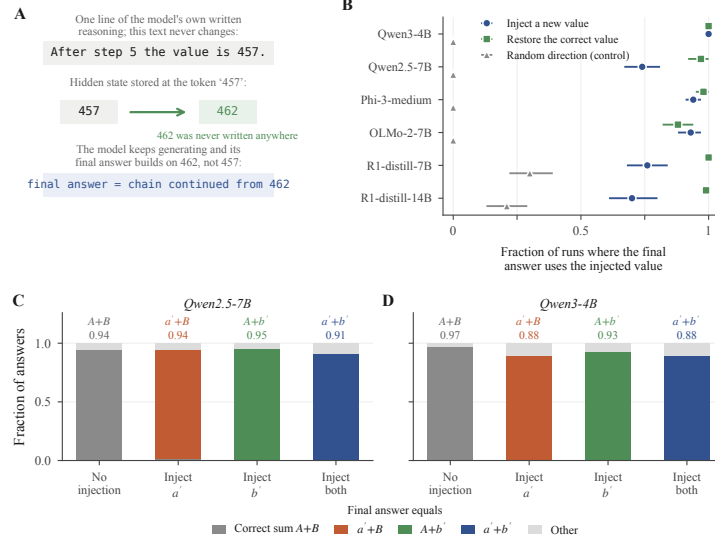


Figure 3: The hidden state under a written number carries the value the model later uses, and two planted values compose. (A) The text still says 457; the hidden state at that token now encodes 462, which appears nowhere, and the final answer continues from 462. (B) Fraction of runs whose final answer uses the injected value; bars are 95 percent bootstrap intervals. The reasoning-distilled models re-solve after their reasoning phase, raising their random-control rate; re-derivation can only produce the correct answer. (C, D) With two counters A and B , injecting a replacement for either or both (a' , b') makes the answer equal the matching sum; answers using only one of two injected values occur at 0.01.

98 generation (next-token agreement 0.87, perplexity ratio 1.12), leaves chain-of-thought accuracy,
 99 bare-answer accuracy, and the amount written unchanged on synthetic chains and on GSM8K, do-
 100 ing no more than a size-matched random-directions ablation. A dose sweep extends the null past the
 101 coherence gate: from a quarter of the calibrated strength to full projection removal (α from 0.125
 102 to 1.0 at $k = 16$, layers 9 to 14 on Qwen2.5-7B), accuracy stays 0.99 to 1.00 in both conditions
 103 and all three arms at one- and two-step chains ($n = 80$ per cell), so no dose separates the J-space
 104 ablation from clean or from the random control. The chain-of-thought-versus-direct ablation asym-
 105 metry reported for J-space [Gurnee et al., 2026] therefore does not replicate on our tasks at any
 106 tested strength, including the one- to two-step depths where models do perform the arithmetic inter-
 107 nally. The null bounds this readout rather than the workspace as a whole (Section 8); Section 4 gives
 108 the gap an existence proof: a value planted in the hidden state under a written number is causally
 109 decisive, sits in residual states the lens samples, and survives removal of the top 16 concepts.

110 4 Written tokens store causally active intermediate values

111 4.1 Restoring and replacing stored values

112 Figure 3 decomposes the effect of editing a written value. Restoring the correct hidden state while
 113 the text still shows a corrupted value returns the answer to correct in 0.88 to 1.00 of cases. Planting
 114 a third value that appears nowhere in the text redirects the answer to it in 0.70 to 1.00 of cases. The
 115 size-matched random control sits at or below 0.01 for the four instruction-tuned models. The patch
 116 replaces the position’s full contextual state rather than an isolated numeric component; because
 117 answers track whichever value’s state is planted, the value that state carries is the causal quantity,
 118 and because the planted value is a mid-chain intermediate the model never produced, the patch does
 119 not inject the answer. The effect persists to 40-step chains (0.95 to 0.98 on Qwen3-4B). Rates are
 120 conditioned on items where the text edit was followed (0.44 to 0.98 by model; Appendix A).

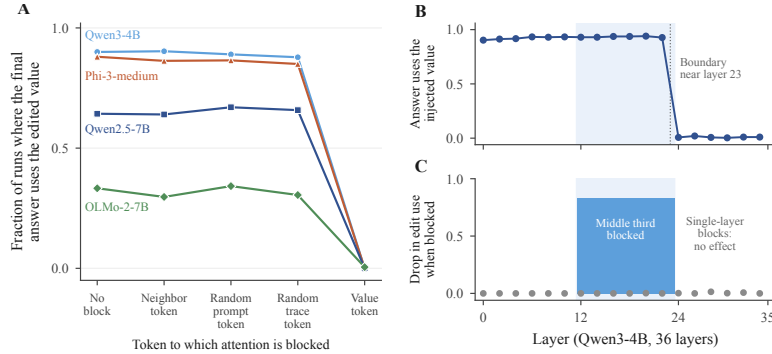


Figure 4: Written values are retrieved through attention to their token, across the middle third of layers. (A) Blocking later positions’ attention to the edited value’s token drops edit use to zero on four models; matched control blocks change nothing. (B) Overwriting the hidden state at any single layer before the read redirects the answer; after it, nothing (Qwen3-4B). (C) Blocking attention across the shaded middle band removes most edit use; no single layer’s block matters (gray dots).

On the narrative task the pattern replicates: a planted never-written count is followed at 0.83 [0.75, 0.90] on Qwen3-4B and 0.86 [0.79, 0.92] on Phi-3-medium, restoring the correct state returns the answer at 0.89 and 0.94, and the random control is 0.00 in both ($n = 86$ and 89).

4.2 Independent storage and composition

In a task with two counters summed at the end (Figure 3C, D), planting a never-written value at one final position yields the corresponding mixed sum (0.94 and 0.95 on Qwen2.5-7B, 0.88 and 0.93 on Qwen3-4B). Planting both yields their joint sum at 0.90 and 0.88, at or above the product of the single-position rates (0.89 and 0.82). The random patch leaves the clean answer in place (0.96 and 0.95), and answers using only one of the two planted values occur at 0.01. The reported number must be assembled at answer time from two values set independently and never written anywhere.

5 How models retrieve written state

5.1 Attention-blocking experiments

Blocking attention from all post-edit positions to the edited value’s position collapses the answer’s dependence on that value on every model tested (Figure 4A); matched blocks of the adjacent words, a random prompt position, or a random trace position leave it at the unblocked level. Blocking the operand the next step needs breaks the arithmetic itself, a positive control showing the mask disrupts computation it targets. The collapse holds across three families and both task formats (natural-language: 0.730 to 0.007 and 0.720 to 0.003 on Qwen3-4B and Phi-3-medium). Under the block the model produces a fluent, parseable number matching neither value (unparseable rate 0.000; OLMo-2-7B 0.040, its unblocked level): it computes forward from state it can no longer retrieve. Reading is the preferred path rather than the only one: with the value’s characters replaced by unreadable dots and attention untouched, the Qwen models reconstruct the value from the previous line in 0.60 to 0.73 of cases, Phi-3-medium never.

5.2 Where retrieval occurs across layers

Layer-resolved versions of both interventions locate the read in depth on Qwen3-4B (36 layers; Figure 4B, C). Overwriting the value’s hidden state at any single layer up to 22 redirects the answer at 0.90 to 0.94; at layer 24 and beyond, at 0.02 or less ($n = 60$): the value can be rewritten at any layer before the read and at none after. Blocking the value-position attention only at the middle third of layers (12 through 23) collapses following from 0.857 to 0.025, blocking only the early or late third leaves it unchanged, and no single layer’s block matters ($n = 80$): the read is distributed redundantly across the middle third and confined to it; the two sweeps agree on the same boundary,

near layer 23. Coarse sweeps on two further models (Appendix B) show the same structure with model-specific depth.

6 When models retrieve versus recompute

By default, models reuse the value they wrote rather than recompute it. On DeepSeek V3.2, an edited value the prompt merely implies is followed at 0.93, while an edited value whose derivation the prompt states is reverted at 0.43; with the defining sentence present but the edit two steps away, following returns to 0.96, so the reversion targets the defined value. Claude Sonnet 4.5 recomputes more broadly. Llama 3.3 70B follows edits at 0.97 either way, despite solving the same chains from scratch: the capacity to recompute does not imply recomputing. Qwen models at 4B and 7B never revert. The conditions are separate experiments; we claim within-experiment contrasts only.

On GSM8K worked solutions, an edited intermediate is followed at 0.87 by Qwen3-4B and at 0.10 by V3.2: a benchmark intermediate is often one operation from numbers still in the prompt. Within one model, V3.2 follows edits at 0.93 on chains it cannot regenerate and at 0.10 on GSM8K. Lanham et al. [2023] report reliance on written reasoning declining with scale on standard benchmarks; our contrasts offer a candidate explanation, that the direction depends on the state’s regenerability. On regenerable state, stronger models depend less on the trace; on state no model can regenerate, dependence is high at every scale. Trace reliance here is a separate quantity from verbalization faithfulness, whether the trace reports the computation that produced the answer; our claims concern reliance on already-written state.

7 Related work

Our interventions build on causal tracing and activation patching [Meng et al., 2022] and causal abstraction [Geiger et al., 2021]; Heimersheim and Nanda [2024] caution that broad patches conflate a component mattering with deleting information at a position, which the single-layer sweep in Section 5 addresses. Faithfulness studies [Turpin et al., 2023, Lanham et al., 2023] and filler-token results [Pfau et al., 2024] bound our claims to state the model does not regenerate internally. Wang [2026] corrupt surface traces only in aggregate and conclude reasoning is latent; our token-level edits show written state is causally decisive when the model cannot regenerate it, which bounds that conclusion. Korbak et al. [2025] argue chain-of-thought monitoring is a useful and fragile oversight channel; our results identify what such a monitor reads and what it cannot.

8 Limitations

Tasks are numeric and count state tracking, so our claims concern serial task state; code, plans, and richer multi-entity state are untested. The J-space null bounds one fitted lens: the full-strength dose sweep covers six middle layers of one model on synthetic chains, we do not measure the J-space content of the written positions directly, and the workspace exceeds the lens’s domain by construction (attention and MLP interiors, the KV cache, and residual directions outside the fitted dictionary). The fine layer sweeps cover one model, and the coarse sweeps do not resolve Qwen2.5-7B’s read boundary. White-box evidence is at 4B to 14B scale, frontier evidence behavioral. Conditioning on edit-following makes cross-model levels not directly comparable; Appendix A reports the conditioning rates. A structured patch control planting a valid activation for a different quantity would isolate value content beyond the size-matched noise control used here.

9 Conclusion

Denied writing, models fail within one to two dependent steps at every scale tested, and no intervention shifts state between workspace and trace: the chain of thought is working memory, settable by intervention, retrieved through middle-layer attention, independently addressable, and decisive for the answer. The divergences between text and stored value shown here are experimenter-induced; the results establish that the representational room for divergence exists and that a monitor reading only the text cannot detect it [Korbak et al., 2025]. Whether it arises naturally is untested.

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A Per-model edit and patch rates

Table 1 gives the per-model rates behind Figure 3B, with the text-edit-following rate that conditions the patch columns. Ten-step chains; brackets are 95 percent bootstrap intervals; n counts items where the text edit was followed. Unconditioned per-run outcomes are in the released data. The distill models re-solve the problem after their reasoning phase, which elevates their random-control reversion: in a text-logged sample, 24 of 24 reverting random-control continuations re-derive from the prompt’s starting value after the reasoning phase, and none correct the value in place. Re-solving can only produce the correct answer, so the planted-value rates are unaffected. R1-distill-1.5B is excluded (12 usable items; random control exceeds the planted-value rate).

Table 1: Text-edit following and state-patch outcomes by model.

Model	Text edit followed	Restore correct state	Plant never-written value	Random control	n
Qwen3-4B	0.98 [0.95, 1.00]	1.00 [1.00, 1.00]	1.00 [0.99, 1.00]	0.00 [0.00, 0.00]	106
Qwen2.5-7B	0.85 [0.80, 0.90]	0.97 [0.92, 1.00]	0.74 [0.67, 0.81]	0.00 [0.00, 0.01]	93
Phi-3-medium	0.94 [0.90, 0.98]	0.98 [0.95, 1.00]	0.94 [0.91, 0.97]	0.00 [0.00, 0.00]	102
OLMo-2-7B	0.85 [0.79, 0.91]	0.88 [0.82, 0.94]	0.93 [0.88, 0.97]	0.00 [0.00, 0.00]	100
R1-distill-7B	0.60 [0.53, 0.66]	1.00 [0.99, 1.00]	0.76 [0.68, 0.84]	0.30 [0.22, 0.39]	65
R1-distill-14B	0.44 [0.36, 0.52]	0.99 [0.98, 1.00]	0.70 [0.61, 0.80]	0.21 [0.13, 0.29]	46

B Coarse cross-model layer sweeps

The state patch applied to one third of layers at a time ($n = 60$ items per model; fraction of runs following the planted value). Qwen3-4B (36 layers): early 0.91, middle 0.93, late 0.00, all layers 0.92. Phi-3-medium (40 layers): early 0.89, middle 0.87, late 0.01, all layers 0.89. Both match the fine sweep’s structure: the planted value takes effect at any band before the read and at none after. Qwen2.5-7B (28 layers) follows the planted value at 0.70 to 0.73 in every band, so its read boundary is not resolved at this granularity.

240 C Generation and intervention configuration

241 Generation uses temperature 0.6, top- p 0.95, five samples per item. The text edit changes only the
242 visible token. The state patch overwrites the residual stream at the value’s token position across a
243 middle band of layers with activations captured from a run in which the different value was written
244 there. The attention block masks all attention from post-edit positions to the value’s token at all
245 layers. Sample sizes are 100 to 600 per condition with two to three seeds. The filler condition
246 prefills the assistant turn with twelve inert dot tokens per dependent step, then decodes the answer
247 at temperature 0. The J-space dose sweep runs the ablation of Section 3 at $k = 16$ and $\alpha \in$
248 $\{0.125, 0.25, 0.5, 0.75, 1.0\}$ with identical items across arms. Prompts, model revisions, and per-
249 run configuration files are released with the code and data.